

E-LEARNING PLATFORM TO HEARING IMPAIRED STUDENTS

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Project Proposal Report

T. N. Samaraweera

B.Sc. (Hons) Degree in Information Technology Specializing in
Data Science

Department of Information Technology
Sri Lanka

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DECLARATION

I hereby declare that the research proposal is my original work and has been prepared by me. The information and data presented in this report are true and accurate to the best of my knowledge. All sources of information and literature used in this report have been properly cited and acknowledged.

I further declare that this proposal has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution, and it complies with the ethical standards of research as outlined by Sri Lanka Institute of Information Technology.

Name	Student ID	Signature
T. N. Samaraweera	IT22143754	

The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of supervisor:-

Date :-

Signature of co-supervisor:-

Date :-

ABSTRACT

AI-driven tools and Large Language Models (LLMs) transformed the nature of education into a personal and accessible experience that cuts across various user groups. Nevertheless, there have been complaints that hearing-impaired students cannot enjoy their studies in most conventional e-learning because the technology does not fully facilitate sign language, emotions, and multi-sensory communication. It requires continued improvements to result in inclusivity, flexibility and in-time responsiveness regarding the areas of specialization in teaching and learning.

The offered system goes further in e-learning exercises as it designs a comprehensive interface that will fit in hearing-impaired students. With a combination of sign language video generation, emotion tracking, multi-sensory immersion and lip-reading assistance, aggregated into a coherent system, it is expected to increase the accessibility and interest in taking care of learning queries and tasks. Such a form of engagement not only adds more depth and personalization to learning experiences but also circumvents short-comings inherent to more traditional platforms when scaled to hearing-impaired learning.

Finally, as shown in this study, this can close the gap between universal e-learning solutions and the necessity of culturally acceptable and domain-specific tools in Sri Lanka, leading to more inclusive and accessible uses not only within the education sector.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ASR	Automatic Speech Recognition
CNN	Convolutional Neural Network
DNN	Deep Neural Network
GCN	Graph Convolutional Network
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
LLM	Large Language Model
LRW	Lip Reading in the Wild (dataset)
LRS2/LRS3	Lip Reading Sentences 2 / Lip Reading Sentences 3 (datasets)
ML	Machine Learning
NLP	Natural Language Processing
RNN	Recurrent Neural Network
TPU	Tensor Processing Unit
VSR	Visual Speech Recognition
3D-CNN	Three-Dimensional Convolutional Neural Network
API	Application Programming Interface
UI	User Interface
UX	User Experience
WebRTC	Web Real-Time Communication
HCI	Human-Computer Interaction
F1 Score	Harmonic mean of Precision and Recall
FPS	Frames Per Second

1. INTRODUCTION

1.1. Background of the Research

VSR or reading lips is the ability of recognizing spoken language by using the process of recognizing the visual movement of the mouth, lips, tongue and the facial area. As opposed to audio based Automatic Speech Recognition (ASR) where acoustic features are used, VSR understands the linguistic contents by using visual information and is, therefore, very valuable to the hearing-impaired or in instances where audio is not available or unreliable like in busy workplaces, silent video transcription and security surveillance.

Hand-crafted appearance characteristics were used at the start like shape descriptors, optical flow and Active Appearance Models (AAMs), which have interfered with the problem of speaker variation, the influence of lighting and the masks of the head. The deep learning, especially the Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) enabled us to train end-to-end at the raw video frames to the text. More recently, AV-HuBERT is a self-supervised model and transformer-based architectures have shown good accuracy through modeling patterns in spatiotemporal lips motion even with minimal labelled data.

Big data on resource-rich languages like English and Mandarin (e.g. Lip Reading in the Wild (LRW) and Lip Reading Sentences (LRS2, LRS3) have been used to drive progress in VSR. Those corpora give thousands of hours of annotated visual speech and allow the deep models to be high-capacity. Nonetheless, there is a lack of such datasets in Sinhala, a language of more than 17 million speakers, with its own phonemes and visemes sets that are different to English. Sinhala languages have retroflex sounds and aspirated consonants which are not found in English and therefore it needs direct transfer by English-trained models and this proves ineffective.

Sinhala VSR is suffering with two major problems:

- Data insufficiency at scale- There is no given widely distributed visual speech corpus with transcripts and viseme labels.
- Deployment constraints There is a need to have the assistive tools operating in real time on the popular devices, yet remaining precise in less than perfect circumstances.

There is yet another gap in the design of user feedback. The system of VSR in place is usually able to display only the text version of information and that may not fully assist hearing impaired users. It is possible to enhance understanding, credibility, and interaction through integration of 3D avatar responses in which a virtual agent animates acknowledged speech. This demands a phoneme viseme mapping using Sinhala specific method and the ease of Lip animation.

The specified gaps would be solved by creating a real-time Sinhala VSR assistant with 3D avatar feedback, which would increase communication capabilities among hearing-impaired individuals and become useful in educational, media, and security fields, as well as in environments with heavy industrial noise levels.

1.2. Literature Survey

Visual Speech Recognition (VSR) also known as lip-reading, is the ability to recognize words from the lips and facial gestures alone without the use of sound. Initial VSR systems were based on hand-designed features such as Active Appearance Models (AAMs) and Discrete Cosine Transform (DCT) coefficients and applied with Hidden Markov Models (HMMs). These were successful in standardized conditions but failed to deal with the variability in real-life conditions on lighting, pose, and manner in which people talk.

Deep learning technology has changed the accuracy of VSR. The recent systems use 3D Convolutional Neural Networks (3D-CNNs) to extract spatio-temporal features of lip movements, but also adopt sequence-model based decoding, such as Transformer or BiLSTMs. Scale English datasets LRW LRS2 and LRS3 have pulled on these improvements, allowing end-to-end training, as well as high accuracy in unconstrained circumstances.

However, recent advances in self-supervised learning, especially such models as AV-HuBERT and Visual Speech Transformers have demonstrated high results even with small amounts of labeled data. They are pre-trained, on large unlabeled Nonetheless, it is revealed that a majority of the studies are targeted at English where the development of low-resource languages is not so far along.

In languages with limited resources like Arabic, Hindi, and Turkish, those who study it have used transfer learning where English-trained models were used together with small native datasets and mappings of phonemes and visemes. Such mapping is extremely important because various languages possess distinct articulation patterns, which relates to the shaping of the lips. Regardless of these achievements the Sinhala lacks a publicly available VSR dataset, mapping of visemes and the use of recent adaptation techniques.

The other topic of research is visual feedback through avatar-based assistive tools. Projects such as Baldi have shown that animated talking head can be used to enhance understanding and interest particularly among the hearing impaired users. These systems can however be audio-based and not connected with live VSR. In the case of Sinhala, there is no system that uses real-time lip-reading in combination with avatar feedback, 3D.

1.3. Research Gap

Although the visual speech recognition (VSR) has already matured to a substantial degree when it comes to well-resourced languages like English, Mandarin, and German, there seems to be very little research on developing VSR in Sinhala. Critical assessment of the current body of literature and systems indicates a number of interconnected gaps that the given research aims to deal with.

- Lack of a publicly available dataset of Sinhala VSR

Nearly every model performing highly on VSR requires labeled datasets of large-scale audio-visual inputs, on which it is trained and tested. Datasets such as LRW, LRS2 and LRS3 present tens to hundreds of hours of labeled video data in English translating to robust spatio-temporal features learning. Nevertheless, the equivalent data is not available even in Sinhala both in terms of isolated words and in continuous sentences at the level of lips reading. The available Sinhala corpora of speech are audio only without viseme (visual phoneme) labeling, that is necessary to train a visual-only system. In its absence, model performance is relegated and cross-lingual transfer cannot be done from English easily because of the variation in phonetic inventory and articulation patterns.

- Phoneme viseme mapping hole on language specific level

Substantial phoneme to viseme mapping is necessary to produce quality VSR. Sinhala has distinctive consonants, vowels and diacritics yielding unique lip shapes not directly captured in English sets of visemes. Local research on synchronization of Sinhala lip to animation and speech synthesis has been done although this work has not been pursued synthetically and comprehensively to come up with a Sinhala viseme mapping that is able to be used in recognition activities. This lack of mapping creates some ambiguity in decoding, and restricts the efficacy of the existing multilingual VSR models.

- Scarcity of using state-of-the-art modeling approaches to low-resource environments

Recent advances (AV-HuBERT and Visual Speech Transformers), demonstrated that even modest self-supervised pre-training can enable a huge reduction in the number of labeled examples that are required to attain high performance. All these methods have found application in the analysis of some languages that have videos data, although there is no documented effort at adapting them to Sinhala. Moreover, low-resource-friendly optimization techniques (e.g. multilingual pre-training and specific (tight) fine-tuning) have not been applied to Sinhala VSR.

- Lack of user centric assistive applications in real time

The majority of the research outputs in VSR are still at experimental or benchmarking phase with less scant attention to actual user-facing deployment in real-time. In situations where assistive applications are present they tend to produce nothing but text. No Sinhala solution to date integrates lip-reading identification with 3D avatar able to simulate speech movements that have been identified by lip-reading in real-time. Such gap is related highly due to the aspects of feedback via the avatar which can be more comprehensible to the hearing impaired, allow them to check the output of the system content easier, and benefit in the trust and engagement.

- Inability to put research and accessibility objectives of the society into a congruous system

The emergence of mobile technology and the increased penetration of the internet in Sri Lanka is insufficient to provide maximum orientation of the advanced AI technologies into inclusive communication tools used by the underrepresented linguistic communities. In the Sinhala-speaking settings, current solutions to the hearing impaired are either by use of sign language or by human lips, which cannot be universally recognized and require special training. A Sinhala VSR AI-based system will fill such a gap, yet there is currently no deployment-ready system.

	Research 1	Research 2	Research 3	Research 4	Proposed
Continuous VSR development sinhala-specific continuous lip-reading	X	X	X	X	✓
Sinhala visual speech collection of large size with viseme labeling	X	X	X	X	✓
Web application optimized on standard devices to be implemented in real-time	X	X	X	X	✓
Processing of 3D avatar feedback of lip-reading (lip-sync animation)	X	X	X	✓	✓
Full end to end implementation of hearing impaired Sinhala users	X	X	X	X	✓

Table 1 : Research gap illustration

1.4. Research Problem

More recent developments in Visual Speech Recognition (VSR) have made publicly available datasets (e.g., LRW, LRS2, LRS3) and techniques (e.g., spatio-temporal CNNs, transformers, self-supervised learning, AV-HuBERT) possible to reach high accuracy on well-resourced languages (e.g., English, Mandarin). Nonetheless, the same advances have not been achieved in Sinhala that has a total of over 17 million speakers because there exist no data, resources that cover that language, as well as more system designs that have been personalized to it.

At the time of writing, there was no existing Sinhala visual speech dataset in form of synchronized video, transcripts, and visemes labeling on a large scale. Such data sparseness hinders the capability to model strong models out of scratch and strangles the potentials of direct transfer in possibility via English-trained models. Phoneme to viseme correspondences in Sinhala and English are quite unlike one another, indicating that the available visemes sets are not available in the same form to be recycled without modifications. This leads to the low accuracy of recognition especially among those phonemes that appear similar visually making it less useful in real life situations.

Parallel, implementation into practice is even more problematic. Any assistive Sinhala lip-reading device should operate in real time, meaning that it should produce with minimum latency and with high accuracy. This is particularly important to the hearing-impaired users, who depend on visual information to have quick comprehension during live conversations. Nevertheless, the majority of research prototypes are made to assess them offline and are too complex to run on common devices such as midrange laptops or smartphones.

The other deficit is in user comments and response. The prevailing systems deliver the results in text form only, which might not accommodate effectively the people with hearing impairments as well. The visual representation of speech movements is often the case when it is needed to verify the knowledge and make mistakes in recognition. The suggested 3D avatar feedback component can solve this issue as it is visually animated the detected speech, which supplements the users with not only a text but also a visual cue. Nevertheless, there is no such system available at the moment to Sinhala, and it demands phoneme-to-viseme mapping to be designed with respect to a language, facility of animation control, and synchronization with the recognition engine.

In a bigger picture it is safe to assume a lost opportunity in developing inclusive technology in Sri Lanka without a Sinhala VSR solution. Hearing Deprived people may communicate through manual lip-reading or signing with Sri Lankan Sign Language which is exclusive in skill and needs to share common knowledge among covers Define. This could be represented in a Sinhala visual lip-reading assistant with a visual feedback loop to fill this gap where the tool is scalable, user-friendly and able to support education, healthcare, workplace communication, and public services.

2. OBJECTIVES

2.1. Main Objective

To formulate, create, and test an AI driven Sinhala Visual Speech Recognition (VSR) assistant that can identify what the person is saying based on the movements of his/her lips in real-time, and present an improved understanding of such action through the incorporation of a 3D avatar feedback engine that promotes a better accessibility, particularly to the hearing impaired audience.

2.2. Sub Objectives

I. Dataset Creation

- A Sinhala audio-visual speech database should be collected and curated to contain a wide variety of speakers, lighting differences and angles.
- Byte-code the dataset correctly transcribed in text, and derive a correct phoneme-to-viseme phonetic mapping in Sinhala.

II. Model Development and Adaptation

- Achieve this burgeoning of data through implementing and optimizing a VSR model based on deep learning with transfer learning and self-supervised models (e.g., AV-HuBERT, 3D CNNs, Transformers).
- Develop the model to run in low latency, real-time applications on readily-available commodity hardware.

III. 3D Avatar Feedback mechanic

- Create a 3D character who also attains synchronized lip synchronization according to known sequences of visemes.
- Include the avatar as part of the VSR pipeline to support dual-channel (text + visual) output.

IV. System Integration

- Create an interface that is comfortable to use and integrates video input, live recognition and visual feedback into one natural environment.
- Make it compatible between platforms and thus deploy it on desktop and mobile.

V. Validity and Verification

- Test the system accuracy with regard to Word Error Rate (WER), the accuracy of recognizing visemes, and the latency.
- Perform usability tests with target participants, especially the people with hearing problems, to test understanding, confidence and adequacy.

VI. Scalability and Future enhancements

- Determine how the dataset and model can be expanded so as to cover bigger vocabularies and more complicated sentences.
- Discuss multi-modal Sinhala ASR/VSR systems integration with audio-based speech recognition.

3. METHODOLOGY

In this section, the methodology of conducting the research and the intended framework to construct AI-Powered Sinhala LipReading Assistant with 3D Avatar Feedback are outlined. Since the last deliverable is web-based application, criterion of software engineering will be adhered to in order to produce a robust, simple and quality product. Iterative development process will be pursued whereby after each development iteration, there would be tests, refinement and optimization of the system technology so that the finished product would perform desired in terms of accuracy, speed and usability.

The Sinhala LipReading Assistant proposed will be comprised of four principle areas:

I. VSR Engine

- Captures and reads video data of webcam or a video file uploaded.
- Tracks what is in the lip region and extracts spatio-temporal features to be recognized.
- Lip to Text: Fine-tunes deep learning models (e.g. 3D CNN + Transformer or AV-HuBERT) and adapts them to Sinhala.

II. Module- Phoneme Viseme

- Maps Between phonemes and Sinhala visemes were identified.
- Treats the unusual articulation patterns that are present in Sinhala (retroflex consonants and aspirated speech tones).
- Makes it compatible with 3D animation needs in rendering natural movement of lips on screen.

III. 3D Avatar FeedbackSystem

- Combines with the viseme result of the recognition module.
- Creates a real-time movement of the lips of a virtual character using the uses blendshape or morph target animation technique.
- Gives a visual feedback in addition to text output to make the information more understandable especially to hearing-impaired users.

IV. Web Application interface

- Provides an interactive, seamless and end-user friendly platform.
- Shows live or recorded recognition results together with synchronized avatar animations.
- Provides customization features in form of avatar style, text size and playback controls.

The paragraph approach means that even the research will generate a fully functional, real-time Sinhala lip-reading web application that has been heavily emphasized on the linguistic correctness, user experience, and practical applicability in the settings of Sri Lanka.

3.1. System Architecture Diagram

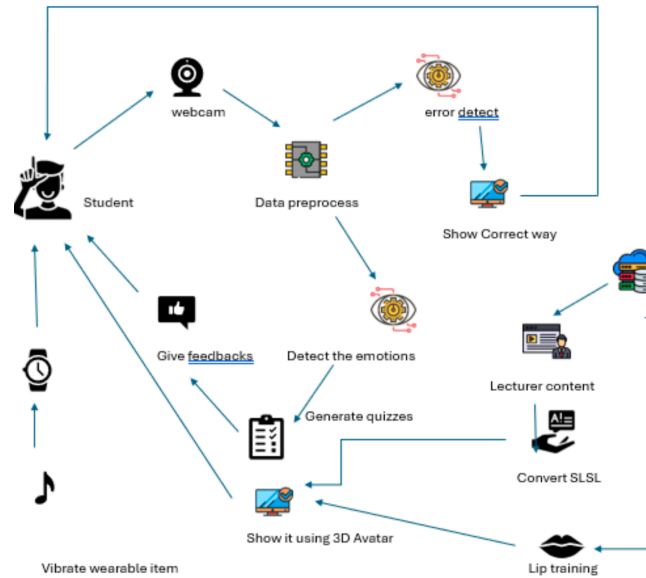


Figure 1: System architecture diagram

The process begins with the User Interface (UI) Module, a web-based dashboard tailored for students and educators. This module captures diverse inputs—webcam data (gestures, emotions, lip movements) and lecturer content (text/audio)—and allows customization of avatar styles (teacher, realistic, cartoon). It serves as the entry point, delivering real-time outputs such as SLSL videos, emotion feedback, haptic cues, and progress reports, fostering an interactive and adaptive learning experience.

Inputs are processed by the Input Processing Module, which employs Natural Language Processing (NLP) for text/audio and computer vision for gesture and emotion analysis. This module segments data into culturally relevant formats, feeding structured outputs (phonemes, gesture features, emotion metrics) to subsequent AI-driven components. The AI-Driven Sign Language Generation Module utilizes Generative Adversarial Networks (GANs), Transformers, and Convolutional Neural Networks (CNNs) to convert processed content into SLSL video sequences. It incorporates facial expressions and non-manual markers, enabling bi-directional communication, and links to the Avatar Customization and 3D Animation Module. This module, powered by Blender and Three.js, animates 3D avatars with customizable styles, ensuring natural and culturally resonant interactions.

The Emotion Recognition with Real-Time Feedback Module leverages CNN/ResNet and reinforcement learning to analyze facial expressions, gaze, and posture, segmenting lessons into 20 minute stages with adaptive prompts and mini-games based on concentration scores. The Multimodal Feedback Mechanism Module enhances engagement by delivering haptic vibrations and visual aids, while the Lip-Reading and Adaptive Learning Module uses CNN-LSTM and phoneme mapping for lip-reading quizzes and individualized pacing. Outputs from these modules converge at the Output Rendering Module, which synchronizes SLSL videos, haptic feedback, and visual dashboards with audio/text, adhering to Universal Design for Learning (UDL) principles and adapting to device capabilities. The rendered outputs feed into the System Output, providing adaptive lessons, feedback, and progress reports, with a feedback loop back to the UI for continuous improvement. The Middleware API Layer, built on Flask/FastAPI, coordinates real-time bi-directional communication (Sinhala-to-SLSL, SLSL-to-Sinhala) across all modules, ensuring low latency and scalability. The Database & Storage Module, implemented with MongoDB, securely stores SLSL datasets, user profiles, emotion logs, and performance data, supporting all components with 99.9% uptime.

3.2. Component specific System Diagram

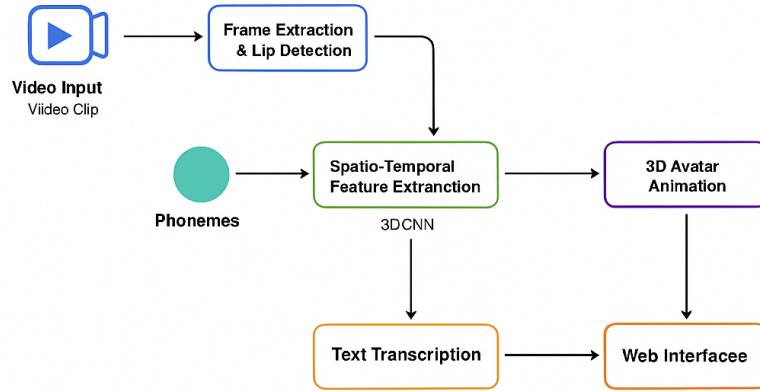


Figure 2: Component specific system diagram

The system chart is based on the component diagram that represents the workflow inside the system under Sinhala **Visual Speech Recognition-based 3D avatar** feedback module that forms the backbone of the AI-powered Sinhala LipReading Assistant.

The steps all start at the **video input step** in which the system can either take in a live video feed through a webcam or a pre-uploaded video by a user. Such raw video data is then fed into the frame extraction and lip detection module, which extracts the mouth area of each frame based on facial landmark-detection algorithm. By taking such step, only the relevant visual data will be allowed to be transferred to the next phase which will enhance accuracy in recognition as well as makes processing less complicated.

The results of the processed frames will then be passed onto the **spatio-temporal feature extraction module** which consists of a 3D Convolutional Neural Network (3D-CNN). This network not only gains the spatial property of the shapes of the lips, but also the temporal component of the lip movement progression over time so that the system can represent the projection of the speech flow visually accurately.

Based on this, the system makes **predictions on phonemes**, which are the smallest units of spoken sound, then they are paired up with their respective visemes in Sinhala. The mapping takes into consideration the peculiar phonetic and aesthetic articulation of the Sinhala speech, with the result being that the avatar is in a position to reflect the mouth movements of the speaker.

The viseme sequence is fed into an animation engine which in turn simulates real-time, natural looking lip sync on the virtual avatar. At the same time, the phoneme sequence through the text transcription module is translated into intelligible Sinhala text output.

Lastly, the animated avatar and the identified Sinhala text can be displayed simultaneously on the web application interface so that the viewers can see them simultaneously. This coordinated multimodal feedback with corresponding visual feedback via the avatar and textual feedback via the transcription increases understanding and the level of active participation by the user especially between the hearing impaired and the user in a noise environment.

3.3. Project Requirements

3.3.1. Functional Requirements

The functional requirements give the concrete actions and characteristics of the system that should be offered to satisfy the expectation of the users. In the case of the Sinhala LipReading Assistant these are:

- I. Video Input Capture
 - Capacity to grab live video on a webcam, or to accept uploaded video files to be processed.
- II. Detection of Lip Region
 - Use detection facial landmarks to isolate the mouth area of the video frames.
- III. Visual Speech Recognition (VSR)
 - Learn spatio-temporal features of lip movement based on deep learning mechanisms.
 - Identify and anticipate Sinhala phonemes based on features of extraction.
- IV. Phoneme–Viseme Mapping
 - To animate correctly, phonemes were mapped to the right ones which were matched to Sinhala visemes.
- V. 3D Avatar animation
 - Use the sequence of visemes to drive an animated virtual avatar as it talks in real time.
 - Have lip movements of the avatar synchronized with identified speech to have the natural visual response.
 -
- VI. Text Transcription
 - Change known phonemes into meaningful Sinhala text indecipherable.
 - Provide the text feedback in the web interface with the animation of the avatar.
- VII. Application Interface on the Web
 - Give an easy, responsive and user-friendly interface.
 - Play avatar animation and text transcription at the same time.
- VIII. Real-time Processing
 - Interactive use requires minimal input to output feedback of video.

3.3.2.Non-Functional Requirements

I. Performance

- Decode video streams and output time-synced at less than 500ms latency so they can be used in real time.

II. Accuracy

- Recognize test datasets with no less than 85 percent phoneme accuracy.

III. Scalability

- Be able to support a large number of users even in concurrency mode.

IV. Compatibility

- Run on popular browsers and devices, and common webcams found in laptops and on required cell phones.

V. Usability

- Give choices that are easy-to-understand and navigate, and provide settings to adjust the display (style of avatar, text font, etc).

VI. Reliability

- Perform within the same standards with minimum decline periods and management of diverse lighting and background conditions.

VII. Security

- Make a safe use of a video input and prohibit the unauthorized access to the data of a user.

VIII. Maintainability

- Permit updating of the recognition engine, avatar module or interface without significant rewrites via modular design.

3.4. Tools and Technologies

3.4.1. Tools

Google Colab:

Google Colab is a cloud-hosted system which enables the user to compose and run python code in the cloud browser without setting up or installing locally. It provides free access to highly computational machines, such as CPUs, GPUs and TPU, that makes it the optimal option in training deep learning versions that target Visual Speech Recognition (VSR). Google Colab, which was created on Jupyter notebooks, allows working with widely-useful AI development libraries, e.g., TensorFlow, PyTorch, and OpenCV, facilitating quick prototyping and experimentation with creating lip-reading and phoneme-viseme mapping algorithms as well as preprocessing kinematic 3D animation data.

Visual Studio Code (VS Code):

The Visual Studio Code is a simple yet mighty code editor that can be further used to create and debug the web application and backend services of the Sinhala LipReading Assistant project. It has a large library of extensions that allow it to integrate with Python, JavaScript and Three.js, all of which are required to use in integrating models and rendering avatars as well as real-time web streaming. VS Code improves the entire stack process, like integrating an AI model and deployment, via its built-in terminal, its Git support, and debugging tools.

Blender:

Blender is a free 3D creation package which could be utilised to create and polish 3D avatar which gives the lip movement feedback address to consumers. It also has support of blendshape/morph target animation, where the lips of the avatar can smoothly and perfectly move in accordance with the identified visemes of the VSR model. Blender has the flexibility that can be used in designing realistic and stylized avatars and this ensures the user engages and gets a clear visual response.

Hugging Face:

Hugging Face enables one to access a wide array of pre-trained deep learning models, including the Transformer-based models that could be used on speech recognition and sequence to sequence translation problems. The models can be enhanced by fitting Transformers library in Sinhala phoneme recognition pipeline and used to generate better models. It is also possible to share trained models and Version control between researchers to future research and collaboration with the Hugging Face Model Hub.

3.4.2. Technologies

3D Convolutional Neural Network (3D-CNN):

The 3D Convolutional Neural Network deals with spatial and temporal characteristics in video examples of lip movement. This type of architecture is highly applicable towards dynamics in the presentations of lips during a given time, allowing proper phoneme recognition within the context of the Sinhala language. The difference between 2D and 3D-CNNs is that unlike 2D CNNs, which obtain single frames and process them independently, 3D have temporal continuity which fits real time lip-reading apps very well.

Transformer sequence model:

Transformers are the new generation sequential data processing architectures that may learn complex dependence between frames. The Transformer model translates the spatio-temporal needs identified by the 3D-CNN into a phoneme sequence in this system, increasing the accuracy of recognition of continuous speech in Sinhala.

Phoneme Viseme Mapping Engine:

The module translates identified phonemes to visemes with regard to the characteristics of Sinhala pronunciation. It warrants that the motion of the lip of the 3D avatar correctly corresponds to the delivered text that facilitates better understanding to the hearing-impaired users.

Three.js:

Three.js is a JavaScript 3D package to render the animated avatar in the web application. It can facilitate live synchronization of the lips and the animation playback is smooth and the avatar feedback works fine in terms of responsiveness on any device.

Python:

Python is mainly used as a programming language on model training, video preprocessing, phoneme prediction, and development of backend service. It is appropriate to use the whole AI pipeline with its simplicity and large library ecosystem, such as OpenCV, Dlib, or PyTorch.

3.5. Implementation

The completed product of the proposed solution is a web-based interactive application intended to identify the Sinhala speech through visual representation (lip-reading) and outputting text-written transcription and synchronous feeds of 3D graphics and the animation of a talking avatar. The application will act as an assistant to hearing-impaired patients and those in a noisy environment so that they can communicate clearly with others in real-time in a visual manner.

Front End Development -

React.js will be utilised to create the Sinhala LipReading Assistant front-end, which will offer a responsive and dynamic interface to enable the reading of live video, real-time text transcription and presentation of 3D avatars. The component-based framework of React enables the modular implementation of the interface components of a live camera stream, avatar view, the transcription box, and the settings section. State management and the virtual DOM help to provide real-time smooth updates and as little as possible delays between the movement of lips being recognized and animation/moving of the avatar.

Back End Development -

This section sits behind the interface and is responsible to generate all the actions, including the approval process, recognition and rewarding. The backend will be established upon **Flask (Python)**, as it is simple, scalable, and deep learning models can be fitted into Flask. Flask will act as the API layer that will process video frames, process the Visual Speech Recognition (VSR) model and output text and viseme sequences that will be used to animate an avatar. Back end will also integrate with AI frameworks like PyTorch or TensorFlow that will perform model inference at low-latency, which can be used in real-time applications.

Implementation of 3D Avatar Animation -

Three.js will be the rendering engine to generate 3D graphics on the browser via the avatar rendering. The backend generated viseme sequence will be loaded onto avatar facial motion as blendshape/morph target animation. This guarantees the adherence to the correct vocalization of the lips to the identified speech. Real-time rendering Avatar models will be generated and tuned in **Blender** optimized to operate in a performance rendering platform.

Database Management -

The application metadata, profiles of the users, language model settings, and performance records will be stored in a **MongoDB** database. Usage data will also be stored in the database in anonymized form in order to assist in refining the accuracy of phoneme-viseme maps used during research. The flexibility of its schema enables MongoDB to store various types of data such as animation parameters, the system settings, and the model versions without affecting functionality.

3.6. Deployment

Hugging Face Spaces will be utilized to deploy the Sinhala LipReading Assistant that will provide a smooth experience of integrating and delivering AI-related applications. This platform allows deploying the trained VSR models and serving them in an interactive interface combining backends inference and frontends presentation.

Deployment will consist of the following:

- **Model Hosting:** On Hugging Face Spaces or another cloud service that supports GPUs, become a host to benefit from hosting model training and experience in practice the benefits of co-location, available to everyone.
- **Web Hosting:** Containerized hosting of the React.js front end and Flask back, in which identical results would be achieved no matter when.
- **Scalability & Access:** The ability of the system to allow multiple users to access and use the system at the same time without any reduction in performance is scalability; the system must be accessible through both desktop and mobile browsers.

The method makes the application reliable, simple to update with easy access by target users including hearing-impaired people and Sinhala listeners living in noisy areas.

3.7. Evaluation of the system

Testing of the Sinhala LipReading Assistant will be conducted using quantitative performance variables like accuracy, speed and qualitative variables like remarks by the users to measure functionality as well as usability.

Precision & Recall:

Precision will quantify the percentage of accurate recognized phonemes and words with regards to the output that was recognized. Recall will determine correct recognitions in terms of recognitions of all real spoken phonemes/words. The expected performance of both metrics is between 70- 80 percent range which keeps the balance between accuracy and coverage within the low-resource language setting.

F1 Score:

The metric to notice is F1 Score which is a combined measure of Precision and Recall, and targeting 75% or further. Such an threshold provides accuracy and reliability of the model with regards to inferring continuous speech recognition in Sinhala.

The precision of Lip Synchronization:

Another specialized metric will be used to measure the degree of correspondence between the lips pattern portrayed by the 3D avatar and actual presence of the Visema pattern followed by the identified speech. The aim is to warrant a synchronization score high enough, say 85% or more; natural and explicable animation of avatars.

User Satisfaction:

The hearing impaired users, the educators, and the general Sinhala speakers will provide feedback by way of surveys and usability tests. It will gauge ease of procedure, the sense of accuracy and usefulness of avatar comments in communications.

For integration of these evaluation measures, the system would be evaluated both on technical performance criteria and practical effectiveness criteria so that the system fulfills the requirement of the concerned individuals as well as the usability and standards in terms of accuracy of the system.

4. WORK BREAKDOWN STRUCTURE

The following is the finalized WBS for the year, developed after extensive brainstorming on the tasks necessary to complete the research. This WBS outlines the key deliverables and associated tasks, breaking down the project into manageable components. By organizing the work into hierarchical levels, it ensures that all critical activities are identified and planned for, facilitating resource allocation, progress tracking, and effective project management. This structured approach provides a clear roadmap for the research, guiding the team toward the successful achievement of the project's goals.

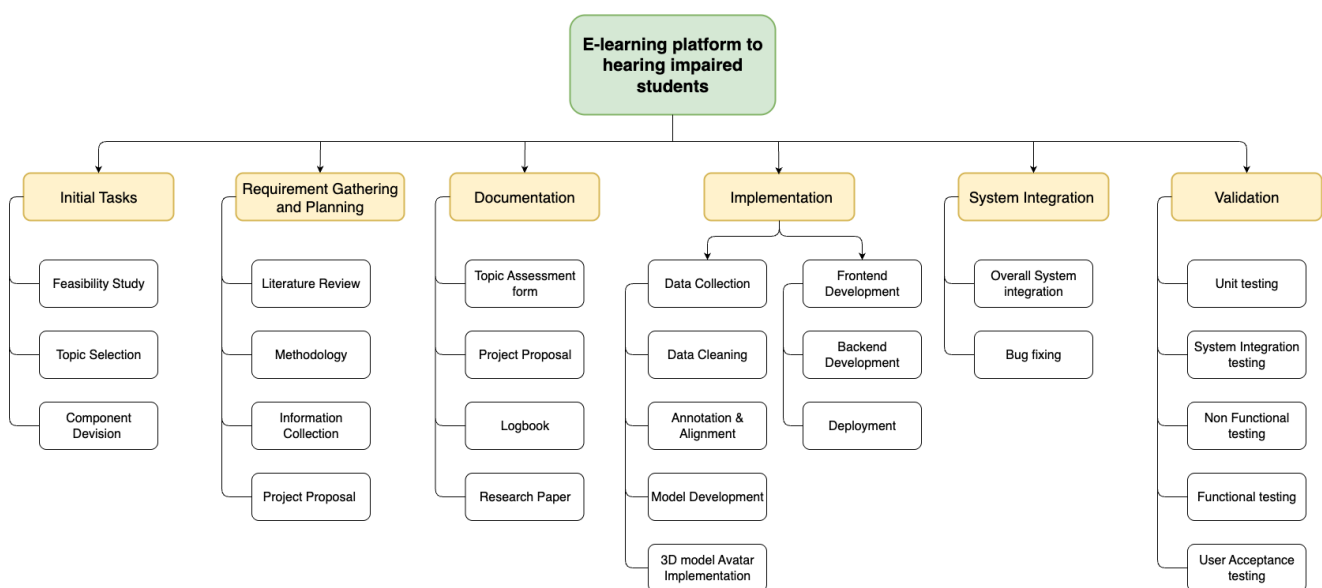


Figure 3: Work breakdown structure

5. GANTT CHART

The Gantt chart below provides an overview of the project timeline we are planning out to work though, outlining the key milestones of the whole project tasks, and the durations for each task respectively. This visual representation will help in tracking progress, managing resources, and ensuring that each phase of the project is completed on schedule.

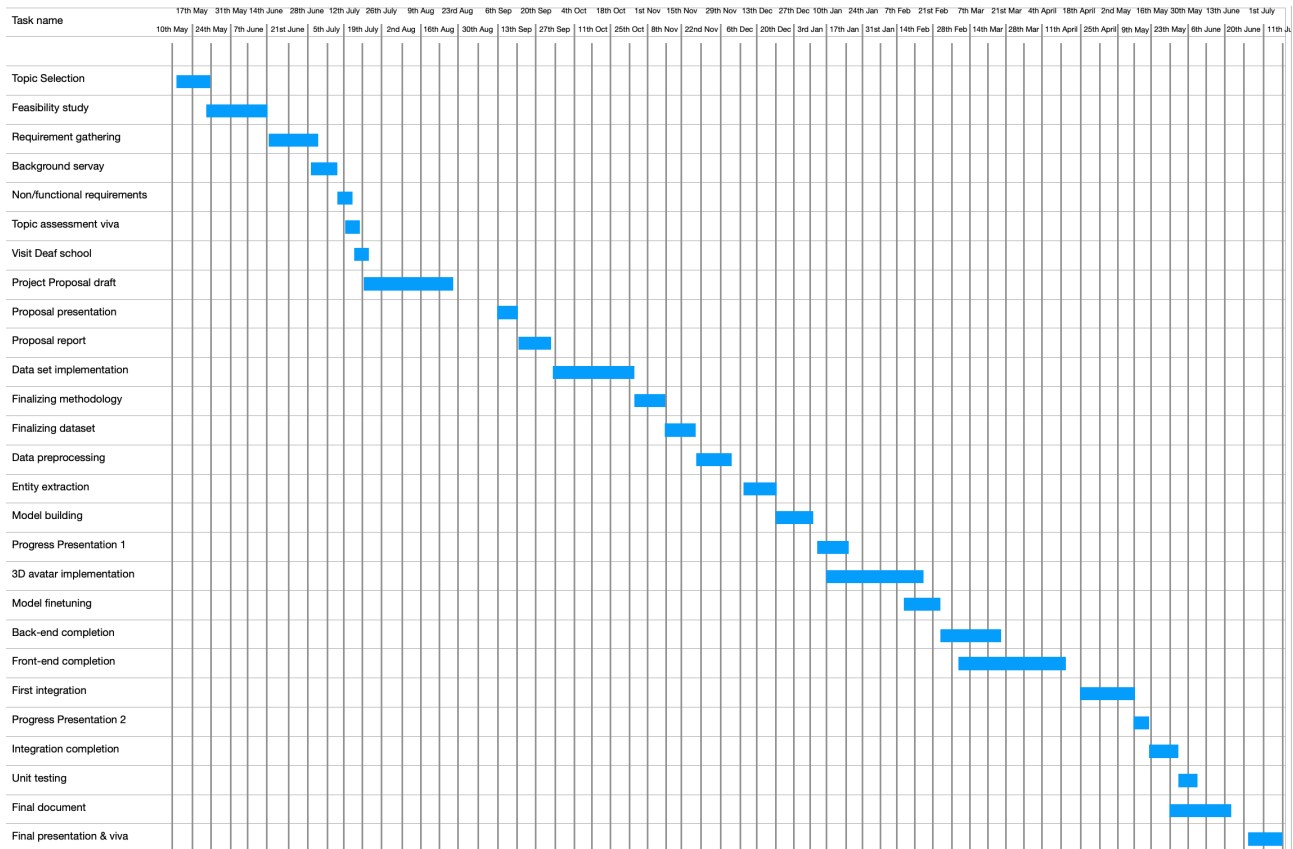


Figure 4: Gantt Chart Diagram

6. COMMERCIALIZATION OF THE PRODUCT

The introduced research project unites several intelligent elements, such as the AI-Powered Sinhala LipReading Assistant with the 3D Avatar Feedback, knowledge-based learning systems, and assistive technologies that will work with high levels of commercialization. The total system influences a wide range of markets- education and healthcare, assistive technology to the differently-abled, AI-based communication systems.

The main channels of commercialization are the implementation of the system as a web and mobile product, and its access to students and educators, as well as other people facing trouble with communication. Joint subscription systems coupled with institutional licensing system in schools, universities and disability training centers sustain the modeling of revenues. Simultaneously, there are some B2B business in other areas, like healthcare (speech therapy, communication between patients), broadcasting (automatic captioning), and government services (airports, courts and customer care centers in noisy places).

Competitive advantages should include this project is localized and inclusive and is aimed towards unrepresented languages, such as Sinhala, and introduced unprecedented features in terms of feedback the use of 3D avatars, which makes it easier to understand than purely text-based solutions. The project can domestic in the region to other languages of South Asia and can also spread to other languages around the world as presenting itself as a scalable, inexpensive, and socially influential AI platform. Additionally, it will appeal to government and non-government organizations to allow the project to be massively adopted.



Figure 5: System Logo

7. BUDGET AND BUDGET JUSTIFICATION

The planned project has well been designed to reduce the budgets with a guarantee of a successful development of an AI-Powered Sinhala LipReading Assistant with 3D Avatar Feedback and the whole integrated system. As it is a research project carried out in a university, the emphasis must be placed on the use of free and open-source tools whenever they are available, spending money only on critical technical infrastructure (storage, hosting, and testing).

Preparation and annotation of datasets use the greatest proportion of the budget (25%), because a high cost of Sinhala lip-movement data of sufficient quality is needed to train and test the system. This involves data storage costs, and data cleaning costs as well as some manual annotation support. Next priority is a **cloud computing and hosting (20%),** which includes the domain registration, server hosting to deploy, and optional subscriptions to stable GPU access during the training of the model as in Google Colab Pro+.

Development tools (15%) consist of low expenses to install specialized plugin or licensing, the vast majority of pieces of software; particularly VS Code, Blender, Unity (free version), TensorFlow, and PyTorch, are open source. **Database and storage (15%),** in turn, is concerned with scale-up options like MongoDB Atlas and Neo4j Aura which have freely available versions that can be augmented with small paid extras of increased storage capacity.

In **testing and evaluation (10%),** there is something set aside as tools for usability testing, accessibility validation tools, multi-device/browser testing to secure the robustness of the system. Lastly, **miscellaneous and contingency allocation (15%)** to help with unexpected technical demands, which might include, related to additional cloud capability or minor third-party libraries.

In general, the proposed budget of 100,000 LKR is realistic and also sustainable in terms of a student research project. It guarantees that all the crucial elements, such as dataset preparation, model training, avatar feedback, and deployment, can be effectively introduced without imposing excessive expenses, and the resulting research value and practical value will be rather high.

Category	Estimated Cost (LKR)
Dataset Preparation & Annotation	25,000
Internet cost	10,000
Cloud Computing & Hosting	20,000
Development Tools	15,000
Database & Storage	15,000
Testing & Evaluation Tools	10,000
Miscellaneous & Contingency	15,000

Table 2 : Budget allocation chart

8. REFERENCE

- I. AV-HuBERT: Shi, B., Hsu, W.-N., Lakhota, K., & Mohamed, A. “Learning Audio-Visual Speech Representation by Masked Multimodal Cluster Prediction.” (AV-HuBERT). arXiv 2022.
- II. VGG Lip Reading datasets (LRW / LRS2 / LRS3). VGG, University of Oxford.
- III. Assael, Y. M., et al. “LipNet: End-to-End Sentence-level Lipreading.” arXiv 2016.
- IV. Qu, L., Weber, C., & Wermter, S. “LipSound2: Self-Supervised Pre-Training for Lip-to-Speech Reconstruction and Lip Reading.” arXiv 2021.
- V. Kim, J. H., et al. “Lip Reading for Low-resource Languages by Learning and Combining General and Language-Specific Knowledge.” ICCV 2023 (paper on low-resource VSR strategies).
- VI. Sinhala viseme mapping research: “Lip Synchronization Modeling for Sinhala Speech” — conference / ResearchGate (Sri Lankan authors).

9. APPENDIX

Below is the similarity score obtained from Turnitin.

The screenshot displays the Turnitin interface for a document titled "E-LEARNING PLATFORM TO HEARING IMPAIRED STUDENTS" (ID: 24J-25-145) by T. N. Samaraweera. The document is a "Project Proposal Report". A red box highlights a match in the text: "B.Sc. (Hons) Degree in Information Technology Specializing in Data Science". The right sidebar shows a "Match Overview" with a total similarity score of 5% and a list of 8 matches.

Match Number	Source	Similarity Percentage
1	Submitted to Sri Lanka ... Student Paper	1%
2	www.coursehero.com Internet Source	1%
3	Submitted to Strathmor... Student Paper	1%
4	Submitted to Munster ... Student Paper	<1%
5	Submitted to Erasmus ... Student Paper	<1%
6	Submitted to South Ba... Student Paper	<1%
7	Submitted to Associati... Student Paper	<1%
8	www.propulsiontechjo... Internet Source	<1%

Figure 6: Turnitin plagiarism check